JC2 H1 Geography Prelims
Mark Scheme

Section A

Cluster 2 Fieldwork

1 With the launch of the Singapore Green Plan, which promotes sustainable living as one of its five pillars, a group of five students wanted to find out the extent to which the CJC community of 1200 people was behaving sustainably. They decided to conduct a study on the use of single-use plastics with the following research question:

To what extent does the CJC community use single-use plastics, such as straws, cutlery and packaging, in the school?

For their investigation, they selected a JC2 recess session where they approached anyone who was willing to respond to their questionnaire. They managed to survey a total of 23 people (18 teachers and 5 students). They also embarked on participant observation one day and stood near the drinks stall to see if students requested for straws. Finally, they went to the rubbish bins in the canteen at the end of the same day to observe the amount of plastic waste disposed. Each member went to a bin to observe using his/her own method of measurement.

Resource 1 shows the map of the school canteen. Resource 2 shows the survey questions used by the students. Resource 3 shows the results of the questionnaire and participant observations.

	(a)	Explain why the students' research question was suitable for the investigation. Award 1 mark for each explanation of a reason why the research question is appropriate. Award a maximum of 1 additional mark for further development of each explanation. Possible Approaches: • The location is located within CJC which is of a suitable scale because it is only 1 school. [1] • This is manageable because of the size of the research group being 5 students that allows them to split up the surveys. [1] • The data needed to answer the research question is researchable because the use of single-use plastics is observable. [1] • Students have access to the community and can gain data needed. [+1] • The aspect of "sustainable living" is clearly defined as the use of single-use plastics. [1] • The type of single-use plastics is also defined. [1] **AO1**	[3]
	(b)	Explain the value of administering a questionnaire and doing participant observations in their research. Award 1 mark for each explanation of value in administering the questionnaire and participant observations. Award a maximum of 1 additional mark for further development of each	[5]

		explanation. A maximum of 3 marks if students only explained for one type of data collection method. Possible Approaches: • The questionnaire could be an efficient way to gather quick responses about the community's behaviour regarding single-use plastics. [1] • The data from the questionnaire can be quantitative, which allows for tabulation of objective data to draw conclusions. [+1] • Based on the data from the questionnaire, students could delve further into some of the areas highlighted by the results of the survey. [1] • Participant observation allows students to observe behaviours/practices that the community might not declare in their responses. [1] • Participant observation allows students to observe a large sample size of staff and students that the questionnaire may not access. [1] • The use of both participant observation and questionnaires can help students corroborate results from different data points. [1] **AO2**	
	(c)	Explain how students can improve their sampling method for the survey. Award 1 mark for each explanation of how students can improve the methods. Award a maximum of 1 additional mark for further development of each explanation. Possible Approaches: • Students can survey JC1 students as well because they make up half of the student community in CJC. [1] • This will increase the reliability of the data. [+1] • They can also increase sample size of the whole community because 23 is not a sizeable representation of the community. [1] • They can also embark on stratified or quota sampling to ensure representation from various groups within the community. [1] • Current representation of 18 teachers and 5 students from JC2 is skewed. [+1] **AO1**	[4]
	(d)	With reference to Resource 2, explain how the students should pilot the questionnaire questions. With reference to Resource 2, explain how the students should pilot the questionnaire questions. Award 1 mark for each explanation of how students can pilot the questions. Award a maximum of 1 additional mark for further development of each explanation. Possible Approaches: • Students can trial the questions with their friends to check for clarity of the questions. [1] • The students could check whether the sequence of questions made sense. [1]	[3]

		• Students can check if respondents understand what "plastics" mean in the survey because it may be understood differently by different people and will result in inaccurate responses. [1] • Students can also check if respondents understand what "frequently" and "rarely" means in the survey as it may be understood differently by different people and will result in inaccurate responses. [1] • Identify a pilot group that is different from the sample group to ensure that the pilot does not influence the responses from the real sample group. [1] • The age range and occupational profile could be similar to the sample in CJC, to ensure similarity with the test group. [1] **AO1**	
	(e)	Using Resource 3, describe the usage of single-use plastics in CJC. Award 1 mark for each description of the usage of single-use plastics in CJC. Award a maximum of 1 additional mark for further development of each analysis. A maximum of 3 marks if not all data types were used. • Most people use single-use plastics either rarely or not very often [1] seen by 20 out of 23 falling into this category. [+1] • All respondents use single-use plastics in CJC. [1] • Most people use a lot of single-use plastics. [1] • 15 out of 23 use "a lot". [+1] • Most people do not ask for or use single-use plastic straws. [1] • 53 vs 20 respondents did not ask for plastic straws. [+1] • Most bins had evidence of high-usage of single-use plastics. [1] **AO2**	[5]
	(f)	Evaluate the students' investigation of the usage of single-use plastics in CJC. Possible Approaches: Candidates could evaluate the investigation by reviewing the methods used to collect data and the data collected in Resource 3. For data collection methods, candidates could review each of the method in turn and/or review all three collectively, including the order in which they were carried out to assess the impact on <u>accuracy and reliability of data</u> collected. Evaluation of these methods could include looking at the small sample size, whether the samples selected are representative of the CJC population, strengths and weaknesses of the different methods, including but not limited to the <u>clarity of phrasing in the survey, the small sample size</u>, <u>the lack of quota sampling,</u> the <u>consistency of methods and definitions of plastic, and the range of data collection methods used.</u> Candidates could consider how such differing interpretations could affect the conclusions that could be drawn from the analysis of these data. *Sample arguments:* *1. The investigation was somewhat valid because it allowed for*	[10]

		behaviour to be investigated via 3 data points. *2. However, the data collected was not representative and hence not reliable to answer the question regarding the whole CJC community.* *3. In addition, the data collected was not always accurate because of the vague survey questions.* *4. Overall, the data was valid enough for a pre-research investigation but not conclusive of the research question.* *Levels marked using Generic Level Descriptors for 10m fieldwork questions* **AO3**	

Section B

Cluster 1 Sustainable Future and Climate Change

Answer **both** questions

2	Resource 4 is a cartogram of world consumption in 2003. Resource 5 shows the changes in the management of all local authority collected waste and recycling rates in England from 2000-01 to 2021/22.		
	(a)	With reference to Resource 4, describe the patterns of world consumption in 2003.	[4]
		Award 1m for each description of world consumption trend. Award a maximum of 1 additional mark for further development of each description, where applicable. Possible responses include: - Developed regions tend to have a higher ecological deficit than less-developed regions. [1] o For example, the majority of countries in Europe experience an ecological deficit while the majority of countries in Africa experiences ecological surplus. [+1] - Countries with highest ecological deficit are in USA, UAE and Kuwait. [1] o These countries have an ecological deficit of 5.4 to 11.0 global hectares per capita. [+1] - Countries with the lowest ecological deficit are in Canada, Brazil, Australia and New Zealand. [1] o These countries have an ecological deficit of -5.4 to -17.8 global hectares per capita. [+1] - Majority of the countries in Asia have moderate levels of ecological deficit of 0.2 global hectares per capita. [1] o Anomaly being Japan and South Korea with higher ecological deficit levels than the rest of Asia. [+1] - Majority of the countries in South America have low levels of ecological deficit. [1] o This ranges between an ecological deficit of -1 to -17.8	

		global hecates per capita. [+1] **AO2**	
	(b)	Cite data from Resource 5 to describe the changes in England's recycling from 2000-01 to 2021/22.	[5]
		Award 1m for each description of changes in England's recycling, with data. Award a maximum of 1 additional mark for further development of each description, where applicable. Possible responses include: - Recycling rates in England has increased from around 10% in 2000/01 to 45% in 2021/2022. [1] - The sharpest increase in recycling rates is from 2000/01 to 2009/10, with an increase of 30%[1] - Recycling rates remain fairly constant from 2009/10 onwards, hovering around 45%. [1] - Recycled-composted amounts have increased from around 3 million tonnes in 2000/01 to 10 million tonnes in 2021/2022. [1] - The increase in recycled-composted amounts were higher in earlier years, increasing from around 3 million tonnes in 2000/01 to 9 million tonnes in 2006/07. [1] - The levels of recycled-composted have remain fairly constant around 10 million tonnes from 2013/14 to 2021/22 [1] **AO2**	
	(c)	With reference to Resource 5, suggest how England's waste management has influenced its progress towards environmental sustainability.	[5]
		Award 1m for each suggestion of how England's waste management has influenced its progress towards environmental sustainability. Award a maximum of 1 additional mark for further development of each suggestion, where applicable. Possible responses include: - The increase in recycling rates and levels have improved England's progress towards environmental sustainability as this is indicative of a shift towards circular metabolism. [1] o As more waste is harnessed as resources from recycling, less waste is being generated due to less need for raw resources for consumption. [+1] - The growth of incineration with EfW has improved England's progress towards environmental sustainability as this may reduce the reliance on the burning of fossil fuels for energy. [1] - However, the rise of incineration, including the slight increase of the use of incineration without EfW may impede environmental sustainability as it releases harmful emissions into the atmosphere, such as particulate matter and greenhouse gases. [1] - The decline in the reliance of landfills has improved England's progress towards environmental sustainability as landfills can cause	

		environmental degradation through pollution. [1] ○ Landfills can pollute groundwater systems through leakage of toxic leachates [+1] ○ Open landfills site may attract pest and bacteria fermentation during decomposition [/+1] - The decline in the reliance of landfills has improved England's progress towards environmental sustainability as landfills release of methane during decomposition, which contributes towards the enhanced greenhouse. [1] **AO2**	
	(d)	Suggest why urban reimaging may result in trade-offs.	[4]
		Award 1m for each suggestion of how urban reimaging may result in trade-offs in sustainable development. Award a maximum of 1 additional mark for further development of each suggestion, where applicable. Possible responses include: - Urban reimaging may result in trade-offs between economic and environmental sustainability as the construction of new buildings to attract investments uses resources and energy. [1] ○ This is especially if urban reimaging adopts a 24-hour city approach, which could result in round-the-clock usage of energy to supply electricity needs. [+1] - The increase in the retail and F&B services available may result in trade-offs between social and environmental sustainability as this may result in higher consumption and more waste being generated. [1] ○ This is especially as urban reimaging increases the number of urban population living in the area, who may have more purchasing power and higher consumption. [+1] - Urban reimaging may also result in trade-offs in social sustainability as the improvement of living standards benefit the high-income groups at the expense of low-income groups when property prices increase [1] **AO1**	
	(e)	Using examples, explain why assessments of liveability are subjective.	[4]
		Award 1m for each explanation using an example of why liveability assessments are subjective. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible responses include: - Liveability assessments may be subjective as the assessment intent would influence the choices of indicators used in the measurement. For example, as EIU and Mercer's assessment are used to calculate hardship allowance, indicators are chosen from the perspective of a well-renumerated expatriate. [1]	

		○ As such, they include indices such as the standard of international schools as a key determinant of the quality of education in the country. [+1] - In contrast, Monocle's assessment is targeted at the wealthy, mobile and cosmopolitan reader, as such the indicators are often more related to culture and lifestyle. [1] ○ For example, Monocle includes metrics such as how late clubs stay open as it prioritises a vibrant nightlife. [+1] - Liveability assessments may also be subjective as they can change over time. The COVID-19 pandemic changed the perception of liveability and wellbeing as greater prioritization was given to effective healthcare and management of the spread of virus. [1] ○ In 2021, Auckland emerged as the most liveable city in EIU's index for the first time due to its good healthcare policies and strict border closures that led to the slowing down of the virus. [+1] ○ In contrast, many European and Canadian cities fell down the rankings, having battled a second Covid-19 wave by restricting cultural and sporting events, and closing schools and restaurants. [+1] **AO1**	

3	Resource 6 is a map of sea surface height change from 1992 to 2019. Resource 7 shows mean global surface temperature anomalies from climate models and observations of surface temperature between 1970 and 2030 using a baseline average from 1900 to 2000. Resource 8 shows a map of the European Union showing percentage of people who think they have to move because of climate change.		
	(a)	Cite data from Resource 6 to describe changes in sea surface height between 1992 and 2019.	[4]
		Award 1m for any of the following. Award 1 additional mark for further development, where appropriate. <u>Possible responses</u> - Sea level rise is highest at about 60 degrees North and South, increasing by up to 15cm. [1] ○ Sea level rise was about 15cm off the Eastern coast of Australia and Japan. [+1] - Sea level also fell at specific areas in the middle of the Atlantic ocean and at the southern end of the Pacific ocean. [1] ○ Sea level fell about 5cm off the coast of Greenland, and up to 15cm off the coast of southern Japan [+] **AO2**	
	(b)	Explain how human actions can lead to trends seen in Resource 7.	[5]
		Award 1m for any of the following. Award 1 additional mark for further development, where appropriate.	

		Possible responses - ~~Temperatures have been increasing since 1990 to 2020 as seen in Resource X [1]~~ - ~~Mean temperature increased after a steep fall from about 0.1degC above average to about 1degC above average. [1]~~ - ~~This increase in temperature could lead to melting of glaciers that flows into the sea, raising sea surface height. [1]~~ - ~~Warmer temperatures could also cause the water to expand, causing sea surfaces to rise. [1]~~ - ~~Sea surfaces fall closer to the poles, and that could be due to temperature differences falling near the poles. [1]~~ - Human actions have increased greenhouse gas emissions resulting in the increase in temperatures that trend upwards, even accounting for 11-year solar cycles (as seen from increasing temperatre anomalies from 1880 to 2000) - (+1) Greenhouse gases trap heat in the lower atmosphere, resulting in increases in surface temperatures. - Human actions such as urbanization have also changed land use, which can increase albedo. - (+1) rapid urbanization and rebuilding from the 1950s onwards have led to a steep climb in temperature anomalies - Land use change has also resulted in fewer carbon sinks, which exacerbates the impact of increased carbon emissions **AO2**	
	(c)	Suggest reasons for the variations shown in Resource 8	[5]
		Award 1m for any of the following. Award 1 additional mark for further development, where appropriate. Possible responses - This could be due to warmer temperatures in the south, resulting in these populations being more exposed to heat stresses. [1] ○ Higher percentage of elderly could lead to more vulnerability to heat stress [+1] - In general, those living more inland tend to have fewer negative sentiments. [1] ○ Austria and Slovakia have just 19% of the population believing they need to move to another region. [+] - This could be due to shelter from sea level rise. [1] - Some places could have better healthcare systems [1] **AO2**	
	(d)	Explain **one** positive impact of climate change.	[2]
		Award 1m for any of the following. Award 1 additional mark for further development, where appropriate. Possible responses	

		- Climate change can extend growing seasons for populations closer to the poles [1] o This can lead to increased productivity of farms [+1] - Climate change can lead to warmer summers resulting in an extended summer tourism season [1] o This can lead to increased tourism activity for countries closer to the poles. [+1] - Climate change can lead to melting of tundra ice, revealing deposits of coal and oil for extraction. [1] o This can allow for economic benefits for countries in the Arctic circle. [+1] **AO1**	
	(e)	Explain why the impact of climate change may be uncertain.	[6]
		Award 1m for any of the following. Award 1 additional mark for further development, where appropriate. <u>Possible responses</u> - There is uncertainty on how much more greenhouse gases will be emitted into the atmosphere [1] o Nature and extent of warming is therefore uncertain. [+1] - There is uncertainty over how much carbon sinks will change over the years. [1] - Warming is also impacted by feedback loops which are not well understood. [1] o Impact of positive feedback loops could result in exponential worsening of the effects of climate change [+1] - Unknown the impact of adaptation on eventual impact of climate change. [1] o E.g. medical advancements or engineering advancements reducing the impact of heat stress [+1] **AO1**	

Section C

Answer **two** questions

4 ‘Meeting the needs of the elderly is essential for successful urban reimaging.’

Evaluate this statement.

[13]

Possible approaches:

Candidates may approach this question by making a judgement on the significance of the elderly in success indicators of urban reimaging. They may

choose to do so through a consideration of criteria such as scale of success, sustainability of success, or impact on individual groups in the city.

Candidates could also approach the question by making a judgment on whether the elderly are necessary for the success of urban reimaging in some contexts more than others through a consideration of two or more case studies where other social groups are present. Candidates could analyse potential trade-offs, which may contribute to the differences in the relative significance of challenges.

Marked using H1 Essay Level Descriptors

AO3

5 Evaluate the success of strategies to improve the lives of slum dwellers in cities.

[13]

Possible approaches:

Candidates could approach the question by making a judgement on whether strategies used to manage housing issues in cities have been successful. They may choose to do this through a consideration of criteria such as the scale of impact, or the nature of impact on stakeholders. Consideration of success should touch on the sustainability of these efforts.

Candidates could also approach the question by making on judgement on whether some types of cities have been more successful at sustainable housing strategies than others through a consideration of two or more case studies.

Marked using H1 Essay Level Descriptors

AO3

6 'Contemporary climate change affects everyone equally.'

[13]

Evaluate this statement.

Possible approaches:

Candidates could approach the question by making a judgement on whether contemporary climate change will affect everyone equally. They may choose to do this through a consideration of criteria such as the scale, extent and irreversibility of impacts. Candidates may also analyse how the impacts of climate change may vary due to differences in physical conditions (such as the uneven changes in temperature and precipitation across places), vulnerabilities of different groups of people (such as uneven vulnerabilities across different ages, gender, and income groups) and adaptive capacity.

Candidates could also approach the question by making on judgement on whether contemporary climate change will affect everyone equally through a consideration of two or more case studies. Candidates could analyse the contextual factors that may contribute towards variations in the impact of contemporary climate change.

Marked using H1 Essay Level Descriptors

AO3